

Gokturk Colak

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EDUCATION

Istanbul University

Bachelor of Arts in Economics, English Program

- **GPA: 3.70/4.00**

Beyazit, Istanbul

Aug. 2022 – Present

University of Tübingen

Erasmus+ Exchange Program – Economics & Business Administration

Tübingen, Germany

Apr. 2026 – Sep. 2026

- Applied Quantitative Methods in Economics & Finance (S310): Monte Carlo inference, MLE, OLS/Logit, and time series analysis in R (see Projects)
- Topics in Crypto (E334): seminar paper on monetary regimes and Bitcoin's fixed supply (see Projects)
- Introduction to Data Science (DS320): completed foundational tutorials in R and Python covering data manipulation and basic analysis

EXPERIENCE

Undergraduate Research Assistant

Istanbul University Informatics Department

Nov. 2023 – June 2024

Vezneciler, Istanbul

- Collected and organized various data sets
- Created complete database of Management Information Systems department heads across 81 Turkish provinces
- Used Microsoft Excel and Google Sheets for data management

PROJECTS

Seminar Paper: Bitcoin's 21M Cap vs. Inflation Targeting

Aug. 2026

- Assessed whether Bitcoin's 21M cap solves the time-inconsistency problem that inflation targeting addresses, building on Kydland–Prescott, Barro–Gordon, and Bernanke–Mishkin
- Compared regimes by their escape clauses using interwar gold standard evidence and Fisher's debt-deflation theory
- Concluded that a fixed supply strengthens commitment but is an inferior monetary rule: debt-deflation risk, extreme price volatility, and a shrinking fee base for network security

Hypothesis Testing, Regression & Time Series | *R, tidyverse, ggplot2*

Aug. 2026

- Simulated type I/II errors and power curves of a one-sided test (1,000 replications) and validated them against asymptotic theory
- Estimated OLS on 17,644 FIFA 23 players and compared LPM vs. Logit (average partial effects, robust SEs) on 45,000 loan applications, diagnosing quasi-complete separation
- Modeled China's CO₂ emissions growth with AR models using ADF tests, AIC/BIC lag selection, and Ljung-Box residual diagnostics

Estimator Properties, MLE & Survival Analysis | *R, tidyverse, ggplot2*

June 2026

- Compared four estimators of a population mean by analytical bias/variance/MSE and Monte Carlo simulation (100 samples, $n = 200$)
- Derived the Bernoulli maximum likelihood estimator, implemented it via numerical optimization (`optim`), and confirmed it equals the sample mean
- Estimated Titanic survival probabilities by class and sex with 95% confidence intervals and discussed why i.i.d. assumptions fail

Algorithmic Trading System Development | *Python, AWS EC2, BinanceAPI*

Aug. 2025 - Dec. 2025

- Signal refinement using an XGBoost Classifier trained on RSI, MACD, and OBV indicators
- Backtesting for investment strategy validation and developing automated market analysis tools
- Hysteresis-based entry/exit logic to maintain momentum-driven positions and filter volatility

Seminar Paper: U.S. Economic Performance, 2008–2023

Feb. 2025 – June 2025

- Analyzed the U.S. economy across four phases (Great Recession, recovery, pre-pandemic expansion, pandemic) using official data sources (FRED, BEA, World Bank, IMF) and 20+ academic reports
- Built a period-by-period indicator table (GDP growth by component, inflation, unemployment, M2, deficits, productivity, sectoral debt); derived financial-sector debt-to-GDP from FRED where IMF data was unavailable
- Evaluated fiscal and monetary policy responses (stimulus acts, Fed rate cuts and QE, TCJA, American Rescue Plan, 2022–23 tightening) against these indicators

TEAMWORK & ACTIVITIES

Istanbul University Young Investors Club (Genç Yatırımcılar Kulübü)

Istanbul, Turkey

Research Department Member

Oct. 2025 – Present

- Conduct research and prepare draft analyses to support the Social Media Department's weekly insights and posts

Istanbul University Data Club

Istanbul, Turkey

Vice President

Oct. 2023 – Oct. 2024

- Organized an institutional visit to IBB Data Laboratory and managed corporate communications
- Volunteered at 22nd Academic Informatics Conference focusing on technology and innovation trends

TECHNICAL SKILLS

Programming Languages: Python, R

Tools: Excel, RStudio, Jupyter Notebook, Git, Github, gretl

Machine Learning: Scikit-Learn, XGBoost, LightGBM, TensorFlow, Keras, caret, tidymodels

Languages: Turkish (Native), English (Advanced), German (Beginner), French (Beginner)

CERTIFICATIONS

Machine Learning for Time Series Data in Python – DataCamp

Exploratory Analysis in R – DataCamp

Next Generation Banking Simulation – Isbank

Digital Banking – Isbank